

LangGraph Pipeline Architecture

MailMind uses a 6-node LangGraph DAG (Directed Acyclic Graph) to process each email. After ingestion and PII masking, nodes 2–4 (triage, commitment, RAG) run in parallel using `asyncio.gather()`, reducing latency by ~60%. Calendar conflict detection and the approval gate run sequentially after the parallel phase.

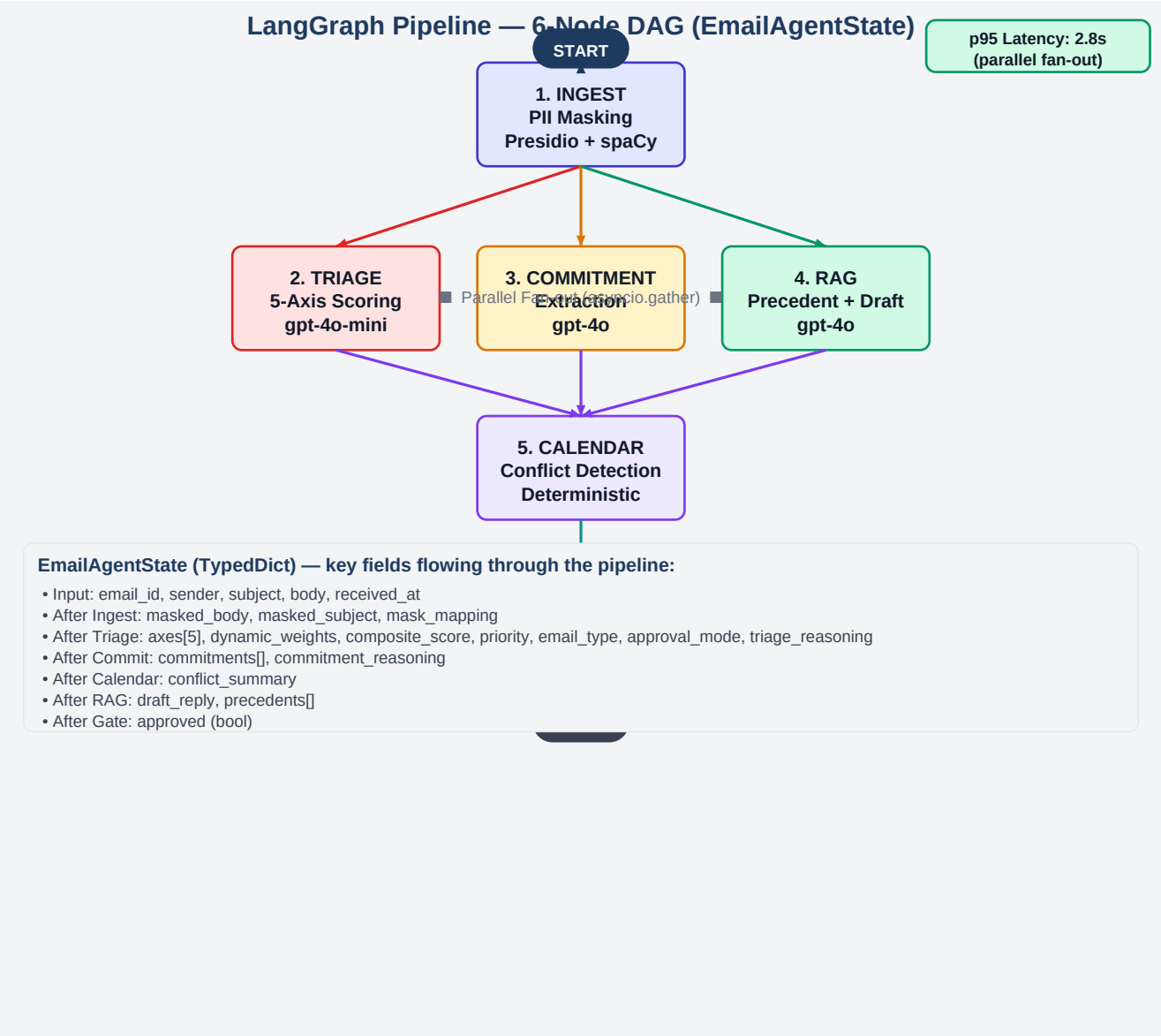


Fig 1: The six pipeline nodes, parallel fan-out edges, and EmailAgentState fields. All nodes share a single TypedDict state object that accumulates results.

OAuth Authentication Flow

MailMind supports Google and Microsoft (Entra ID) OAuth 2.0. After the consent redirect, the backend exchanges the code for tokens, Fernet-encrypts them at rest, upserts the user/account rows, creates a session with SHA-256 hashed token, and returns an HttpOnly cookie (SameSite=Lax).

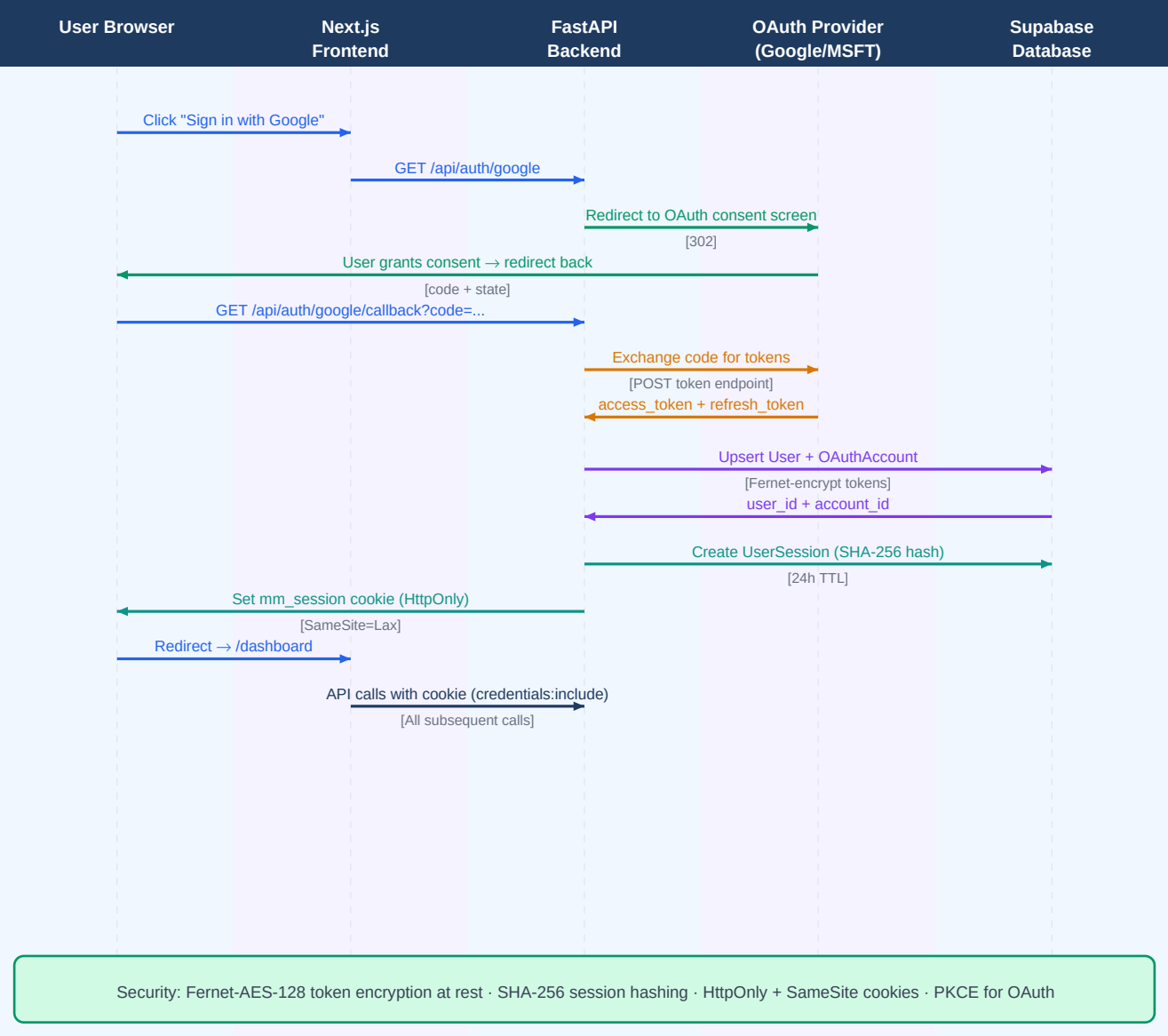
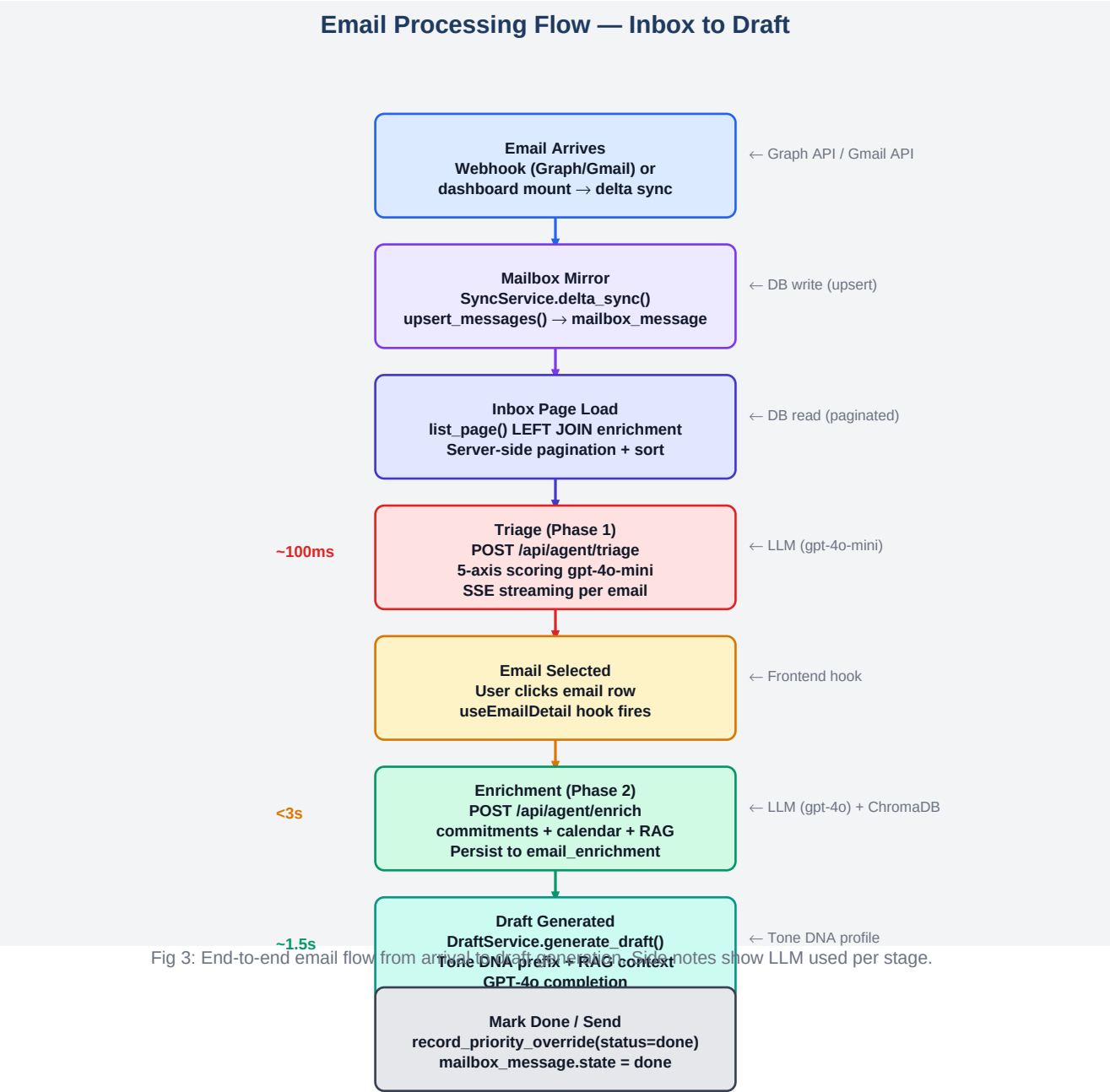


Fig 2: Swim-lane OAuth flow. Tokens encrypted with Fernet-AES-128; sessions hashed with SHA-256.

Email Processing Flow

Email processing is split into two user-facing phases. Phase 1 (triage) runs immediately on inbox load and populates priority badges within ~100ms per email using SSE streaming. Phase 2 (enrichment) runs when an email is opened, fetching commitments, calendar conflicts, RAG precedents, and a draft reply.



Mailbox Sync & Webhook Flow

MailMind maintains a server-side mailbox mirror (MailboxMessage table) synced via Microsoft Graph delta queries and Gmail history.list. Webhooks trigger near-real-time delta syncs. If BACKEND_PUBLIC_URL is unset, the system degrades gracefully to on-mount triggered syncs.

